

# FAST-NU IQ & Analytical Skills 65

## Questions: Practice from Past Papers

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- 4- At the end of every Live Session, please share your feedback and suggestions. Your feedback helps me improve the upcoming sessions and cover the topics according to your needs.
- 5- Tomorrow In Sha' Allah, we will cover Sequence & Series of Advanced Mathematics In Sha' ALLAH.

Stay consistent, keep working hard, and prepare with full focus.

Best of luck to all of you!

Check Point :-

Find the Missing Number

|    |    |    |
|----|----|----|
| 6  | 15 | 23 |
| 4  | ?  | 30 |
| 45 | 41 | 36 |

- A) 30   B) 32   C) 48   D) 50   E) 55

## Section 1: Coding and Decoding

1. If **SOMETHING** stands for 123456789, then **NIGHT** stands for?

- a) 87965
- b) 36789
- c) 37896
- d) 38976
- e) 37895

↑↑↑

SOM ↑↓

↓ ↓ ↓  
8 7 9 6 5

87965

2. If **653042** stands for **VOLUME**, then **4032** stands for?

- a) MEUL
- b) MLUE
- c) MELU
- d) MULE
- e) MLEU

VOLUME  
6 5 3 0 4 2

4 0 3 2  
M U L E

A B C D

S T U

3. If **TRAIN** is coded as **UTDMS**, then **PLANE** will be coded as:

- a) QNDRJ
- b) QNDQJ
- c) OMZKD
- d) QODRJ
- e) QNCQJ

↓ ↓  
T R A I N  
+1 +2 +3 +4 +5  
U T D M S

R, S, T

K L M N

I J K L M

N O P Q R S

PLANE  
Q N D R

4) If CBKPDG stands for

BALUCH, then what stands

nnnrdr?

d) PTDSR2

for ORDERS?

a) PSCFSK b) PSC RSF c) PPT d) PIDSK

B A L O C H      ORDERS  
 $\downarrow +1$      $\downarrow +1$      $\downarrow -1$      $\downarrow +1$      $\downarrow$      $\downarrow$   
 C B K P D G      PSCF

K L M

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| A | B | C | D | E | - |
| 1 | 2 | 3 | 4 | 5 |   |

✓ Reversed Alphabet

→ ←

Q If MATH is coded as NZGS then CODE is coded as? X L W V

will be coded as! X L N V  
 (a) X M N V b) ~~X L V W~~ c) X L N V d) ~~X V L N~~

|     |   |   |   |   |   |              |
|-----|---|---|---|---|---|--------------|
| (A) | B | C | D | E | F | G            |
| Z   | Y | X | W | V | U | T            |
| H   | I | J | K | L | M | N            |
| S   | R | Q | P | O | N | <del>M</del> |

(O) L

← M N U

5) If TIGER is coded as

U H H D S  
 (+1) (-1) (+1) (-1) (+1)

G H I

HORSE

I N S R F

Vowels + Consonants

← T U

Q If TABLE is coded as  
 SCAKG  
 HOUSE?  
 EFGH → ABC FGH  
 (-1) (+2) (-1) (-1) (+2)  
 G Q W R G

TEACHER  
 1 2 3 4 5 2 6

HEART  
 5 2 3 6 1

→ Ist Occur — Look (104)  
 A B C D E . . . W X Y Z  
 1 2 3 4 5 23 24 25 26

Q APPLE ✓ the  
 1 2 2 3 4 PAPER  
 =  
 [1 2 1 3 4]

Q APPLE is —  
 1 2 2 3 4 BANANA  
 1 2 3 2 3 2

Q Alphabet Find is  
 If CAT is 60  
 3 1 20  
 EF 4

DOG  
4 15 7

Q BAD  
2 1 4 tw

FACE  
6 1 3 5

## Section 2: Age Problems

4. A father is 3 times as old as his son. In 10 years he will be twice as old as his son. What is the present age of the father in years?

- a) 28
- b) 36
- c) 30 ✓
- d) 49
- e) None of the above

$$F = 3S$$

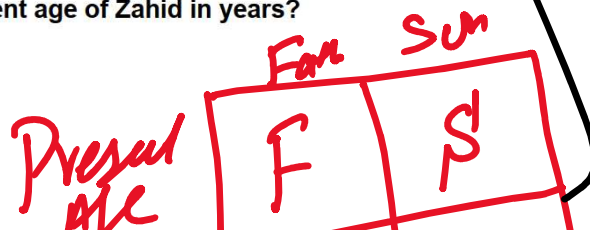
$$F + 10 = 2(S + 10)$$

$$3S + 10 = 2S + 20$$

$$S = 10$$

5. Zahid is 5 years older than his wife, who is 5 times as old as her daughter. If 4 years ago the age of the daughter was 5 years, what is the present age of Zahid in years?

- a) 50
- b) 40
- c) 35
- d) 55
- e) 45



$$M = B + 2$$

- d) 55  
e) 45

| Present age | F    | P    |
|-------------|------|------|
| A 10 yrs    | F+10 | S+10 |

$M = B+2$

$$Z = W + 5 \rightarrow (1)$$

$$W = S + D \rightarrow (2)$$

$D = 9 \text{ yrs}$

$W = 45$

"D" ✓

$D - 4 = 5 \text{ yrs}$

6. Ali is twice as old as Bilal. Five years ago, Ali was three times as old as Bilal. What is Ali's present age?

- a) 20  
b) 25  
c) 30  
d) 35  
e) 15

$$A = 2B \rightarrow (1)$$

$$A - 5 = 3(B - 5)$$

$$2B - 5 = 3B - 15$$

$$B = 10$$

A, B

$\downarrow \quad \downarrow$

A-5 B-5



$$\boxed{B=10}$$

### Section 3: Odd One Out

7. Which word describes something different from the others?

- \* a) Incapable
- b) Clever
- c) Talented
- d) Skilful

8. Identify the odd one.

- a) Saeed Anwar
- b) Inzamam ul Haq
- c) Virat Kohli
- ☒ d) Jahangir Khan
- e) Rohit Sharma

→ Squash

Am (M)

9. Which of the following words would appear as the last word in a dictionary?

- a) Fallow
- b) Fallon
- c) Fallible
- ☒ d) Fanciful
- e) False

Fan

Ant

10. Which one is different from the others?

- a) Mercury
- b) Venus
- c) Mars
- d) Jupiter
- ☒ e) Moon

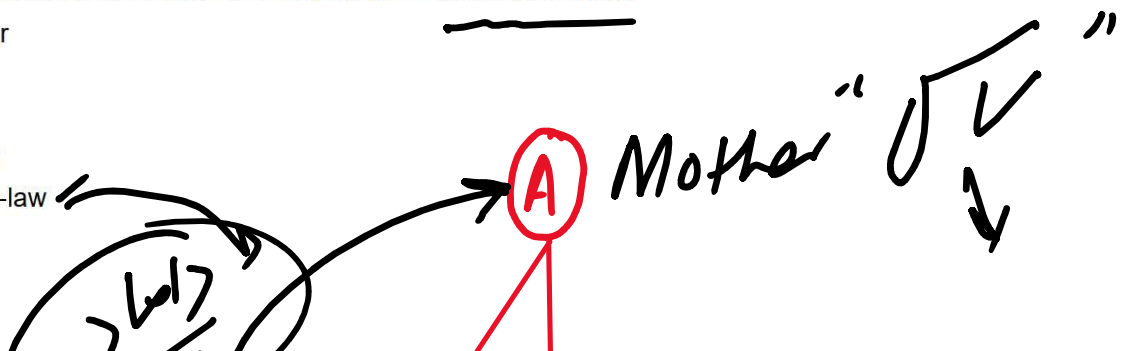
Aunt

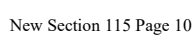
→ Natural

### Section 4: Blood Relations

11. A is mother of B and C. C is wife of D. What is D to A?

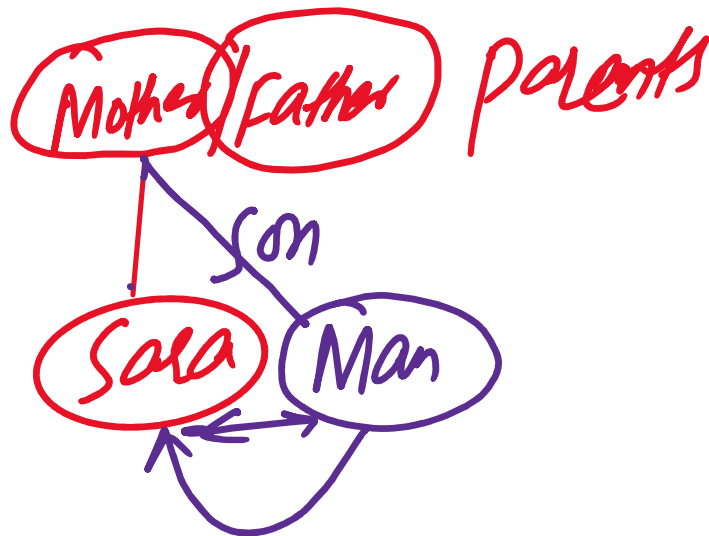
- a) Brother
- b) Son
- c) Uncle
- d) Father
- e) Son-in-law





14. Pointing to a man, Sara said, 'He is the only son of my mother's husband.' How is the man related to Sara?

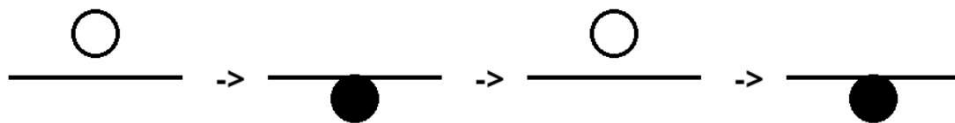
- a) Father
- b) Brother ✓
- c) Uncle
- d) Husband
- e) Son



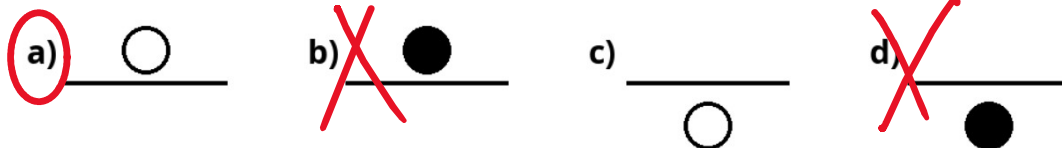
## Section 5: Non-Verbal Reasoning

15. Which figure comes next?

Figure series:

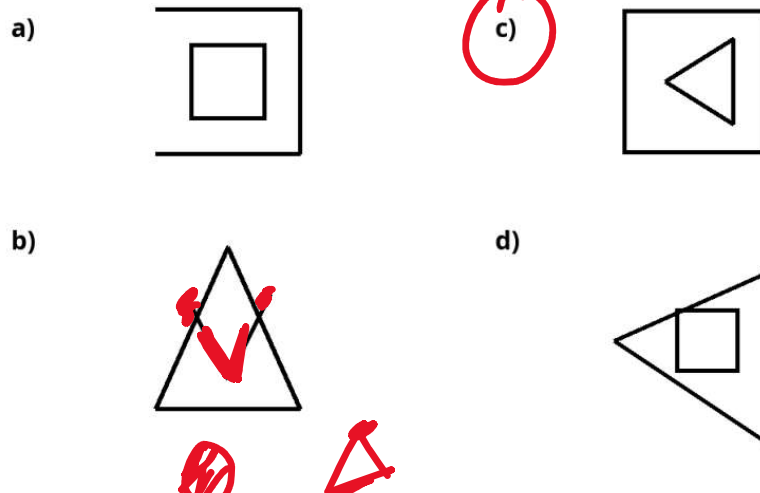


Options:



16. Which is the odd one out?

Odd figure out:



17)



17)

Figure analogy:



- a) b) c) d)

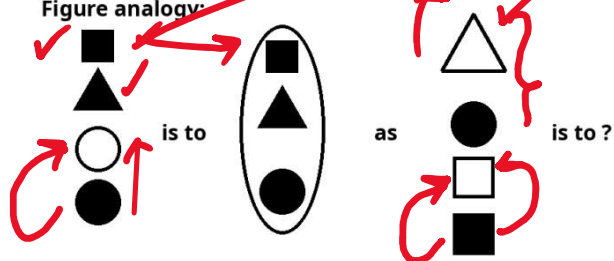
18 Which is the odd one out?

Odd figure out:

- a) b) c) d) e)

19 Complete the figure analogy.

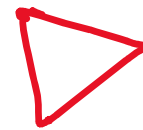
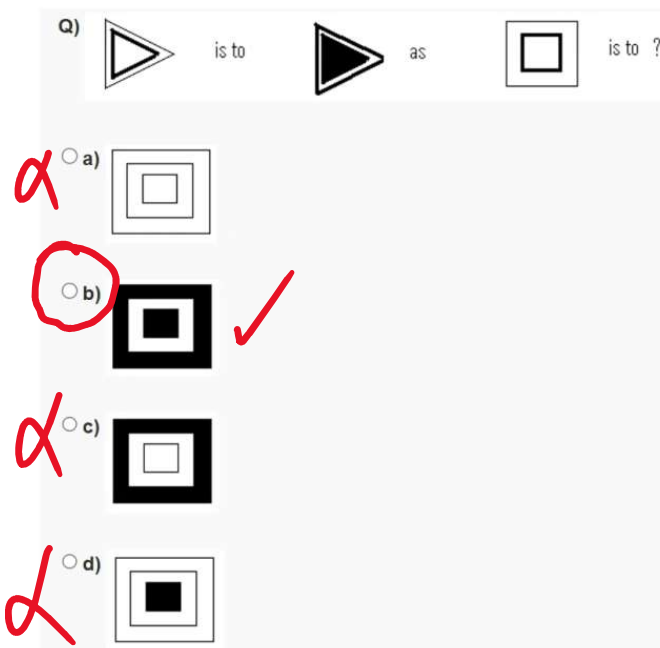
Figure analogy:



- a) b) c) d)

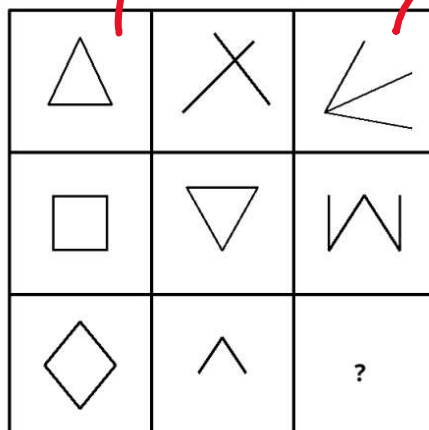
20. Complete the figure analogy.

Figure analogy:



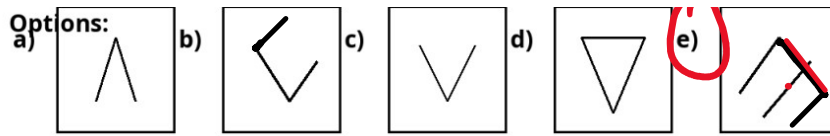
21. The missing square is:

Missing square matrix:



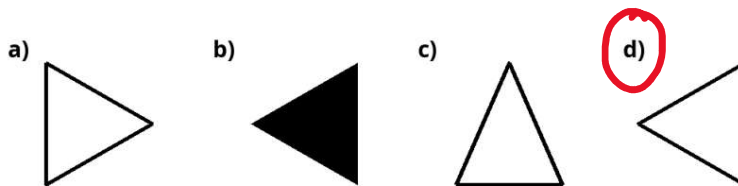
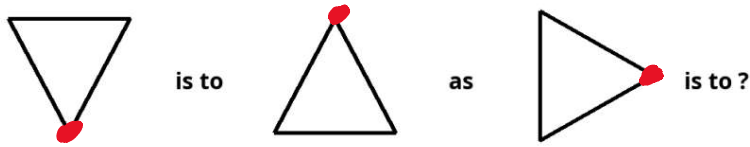
Options:





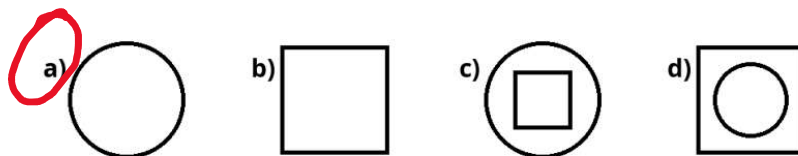
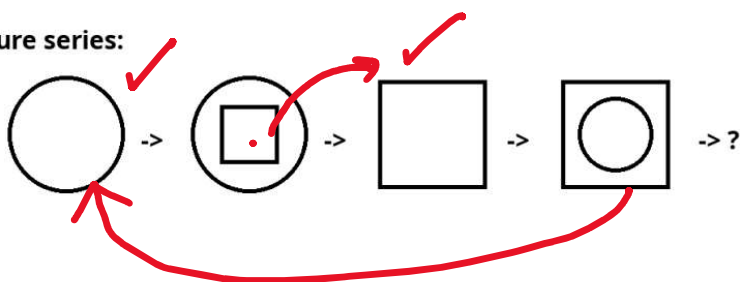
22. Complete the figure analogy.

Figure analogy:



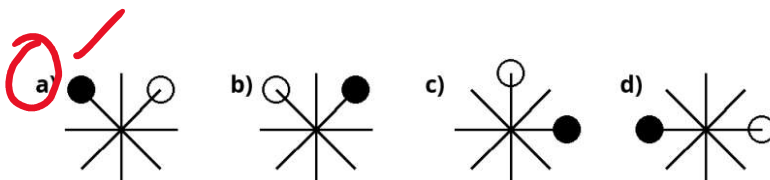
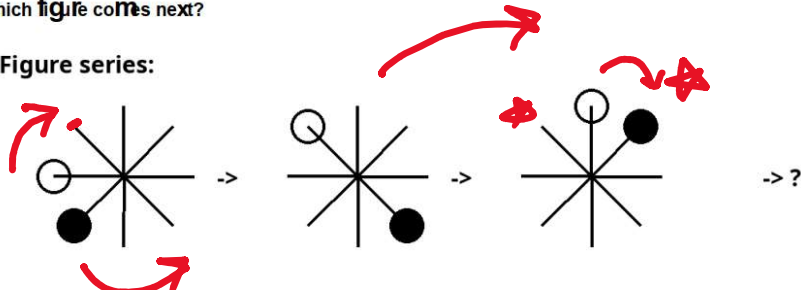
23. Which figure comes next?

Figure series:



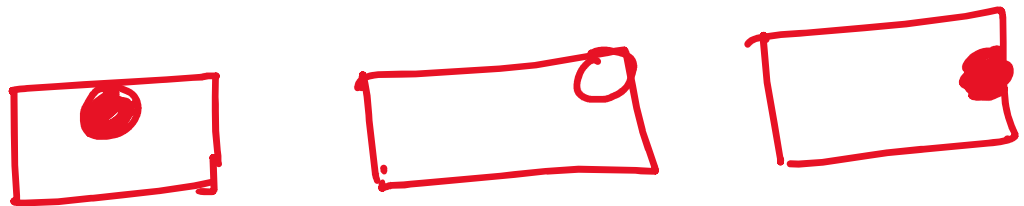
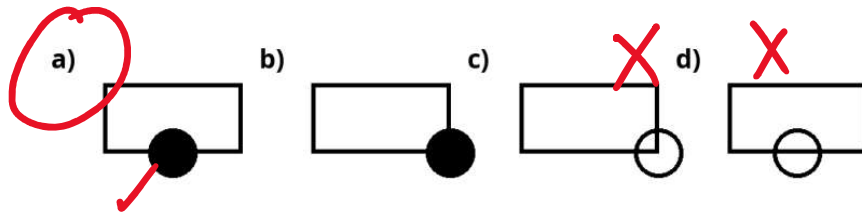
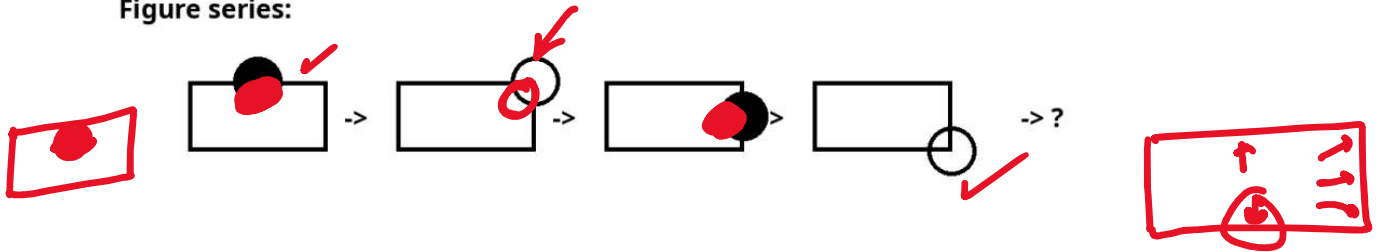
24. Which figure comes next?

Figure series:



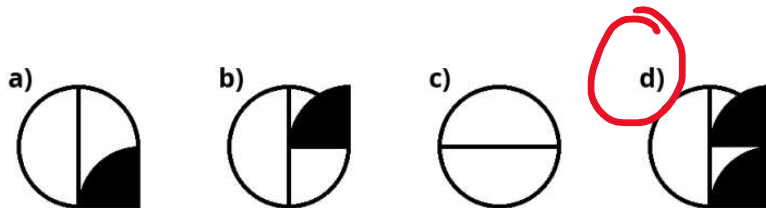
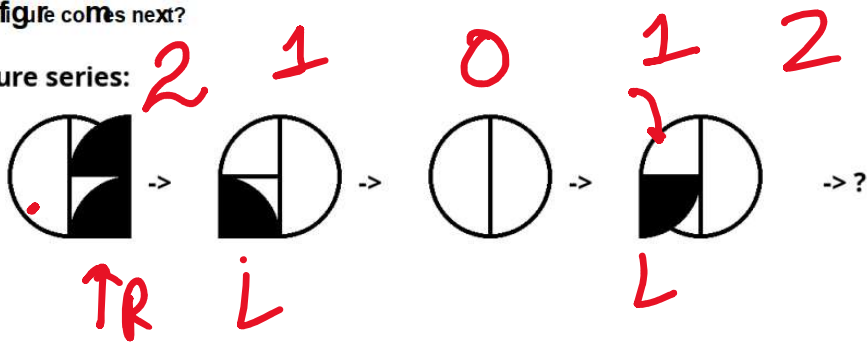
25. Which figure comes next?

Figure series:



26. Which figure comes next?

Figure series:





## Section 6: Logical Reasoning

27. A study of a math program implemented in several pre-schools indicates that children who received specialized Math Plus education between the ages of three and five had significantly higher math scores in 3rd and 4th grade than classmates who did not receive this instruction. The proponents argue that introducing this program for all children aged three to five will significantly improve their chances for success in school. Which of the following, if true, would most seriously weaken the argument?

- a) Children are not really interested in enrichment programs in preschool.
- b) Most parents send their children to preschool for social development and do not have a clear idea about what types of education they want for their children.
- c) The cost factor needs to be specified and established before a large scale program can be undertaken.
- d) Cognitive abilities of 3- to 5-year-old children are constantly changing.
- e) The children in the pre-schools that were studied had previously been exposed to another math enrichment program.

✓ 28. People should be held accountable for their own behavior whether or not it leads to punishment. However, no person should be held accountable for behavior over which he or she had no control. Which of the following is the most logical conclusion of the argument above?

- a) People have control over their own behavior.
- b) People should not be held accountable for the behavior of other people.
- c) Behavior that cannot be controlled should not be punished.
- d) People have control over the behavior of other people.
- e) People cannot control the behavior of other people.

29. The symptoms of mental disorders are behavioral, cognitive, or emotional problems. Some patients with mental disorders can be effectively treated with psychotherapy, but it is now known that in some patients mental disorders result from chemical imbalances affecting the brain. Thus, these patients can be effectively treated only with medication that will reduce or correct the imbalance. The argument depends on assuming which one of the following?

- a) Treatment by psychotherapy can produce no effective reduction in or correction of chemical imbalances that cause mental disorders.
- b) Treatment with medication always shows faster results for patients with mental disorders than does treatment with psychotherapy.
- c) Most mental disorders are not the result of chemical imbalances affecting the brain.
- d) Treatment with psychotherapy has no effect on mental disorders other than a reduction of the symptoms.
- e) Medication is always more effective in treating patients with mental disorders than is psychotherapy.

30. Whitley Hospital's much publicized increase in emergency room efficiency due to its new procedures for handling trauma patients does not withstand careful analysis. The average time before treatment for all patients is nearly 40 minutes - the highest in the city. For trauma victims, who are the specific target of the guidelines, the situation is even worse: the average time before treatment is nearly half an hour - more than twice the city average. Which of the following, if true, would most seriously weaken the conclusion about the value of the new procedures?

- a) The recently hired administrators who instituted the new procedures also increased Whitley's emergency room staff by nearly 15 percent.
- b) After the new procedures went into effect, Whitley's average time before treatment for trauma patients and patients in general dropped by nearly 35 percent.
- c) Due to differences in location and size of staff, not all emergency rooms can be expected to reach similar levels of efficiency.
- d) Because trauma patients account for a large percentage of emergency room patients, procedures that hasten their treatment will likely increase overall emergency room efficiency.
- e) The city hospitals with the most efficient emergency rooms utilize the same procedures for handling trauma patients as does Whitley Hospital.



31. It doesn't surprise me that the critic on our local radio station went off on another tirade today about the city men's choir. This is not the first time that he has criticized the choir. But this time his criticisms were simply inaccurate and unjustified. For ten minutes, he spoke of nothing but the choir's lack of expressiveness. As a professional vocal instructor, I have met with these singers individually; I can state with complete confidence that each member of the choir has quite an expressive voice. Which one of the following is the most serious flaw in the author's reasoning?

- a) He attempts to conclude the truth of a general situation from evidence about one specific situation.
- b) He cites his own professional expertise as the sole explanation for his defense of the choir.
- c) He ignores evidence that the critic's remarks might in fact be justified.
- ☒ d) He assumes that a group will have a given attribute if each of its parts has that attribute.
- e) He directs his argument against the critic's character rather than against his claims.

32. Although most of the fastest growing jobs in today's economy will require a college degree, many of the new jobs being created - from home health aide to desktop publisher - require knowledge other than that gained from earning a degree. For workers in those jobs, good basic skills in reading, communication, and mathematics play an important role in getting a job and developing a career. From the information given above it can be validly concluded that, in today's economy:

- a) A job as a home health aide will rely more on communication skills than on basic skills in reading and mathematics.
- b) Desktop publisher jobs and home health aide jobs are not among the fastest growing jobs.
- c) The majority of the new jobs being created require knowledge other than that gained from earning a college degree.
- d) If a job is one of the fastest growing jobs, it will require a college degree.
- ☒ e) Skills in reading, communication, and mathematics play an important role in developing a career as a desktop publisher.

33. Ever since I arrived at the college last week, I've been shocked by the poor behavior of the students. The student population is completely lacking in proper social skills. Which of the following, if true, would weaken the above conclusion?

- ☒ a) Students who are away from their parents often exhibit rude behavior.
- ☒ b) The college numbers over 50,000 students.
- c) The narrator is a student and has interacted with many students.
- d) The narrator was reluctant to stay at the college.
- e) Social skills should not be expected of college students.

34. Lycopene, glutathione, and glutamine are powerful antioxidants that neutralize the free radicals that are produced in the body as a result of routine bodily processes. An excess of these free radicals in your system causes rapid aging because they accelerate the rate of cellular damage. Aging is simply the result of this damage. Thus, to slow down aging it is necessary to supplement your diet with these antioxidants on a daily basis. Which of the following, if true, most seriously undermines the author's contention?

- a) Smoking cigarettes is one of the main causes of cellular damage in humans.
- b) Most persons aren't concerned with the effects of aging until it is too late to do anything.
- c) Only overweight people who do not exercise on a daily basis are likely to have an excess of free radicals in their systems.
- ☒ d) Exercise associated with normal daily activities effectively neutralizes and dissipates the free radicals that are produced as a result of routine bodily processes.
- e) The cost of antioxidants is exorbitantly high and well beyond the budget of most consumers.

35. There is clear evidence that the mandated use of safety seats by children under age four has resulted in fewer child fatalities over the past five years. Compared to the five-year period prior to the passage of laws requiring the use of safety seats, fatalities of children under age four have decreased by 30 percent. Which one of the following, if true, most substantially strengthens the argument above?

- a) The number of teenage drivers has increased by 30 percent over the past five years.
- ☒ b) The number of serious automobile accidents involving children under age four has remained steady over the past five years.
- c) Most fatal automobile accidents involving children under age four occur in the driveway of their home.
- d) Automobile accidents involving children have decreased sharply over the past five years.



35. There is clear evidence that the mandated use of safety seats by children under age four has resulted in fewer child fatalities over the past five years. Compared to the five-year period prior to the passage of laws requiring the use of safety seats, fatalities of children under age four have decreased by 30 percent. Which one of the following, if true, most substantially strengthens the argument above?

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- c) Most fatal automobile accidents involving children under age four occur in the driveway of their home.
- d) Automobile accidents involving children have decreased sharply over the past five years.
- e) The use of air bags in automobiles has increased by 30 percent over the past five years.

36. Newspaper publishers earn their profits primarily from advertising revenue, and potential advertisers are more likely to advertise in newspapers with a wide circulation - a large number of subscribers and other readers - than with other newspapers. But the circulation of the newspaper that is currently the most profitable one in this city has steadily declined during the last two years, while the circulation of one of its competitors has steadily increased. Any of the following, if true, would help explain the apparent discrepancy between the two statements above EXCEPT:

- a) The most profitable newspaper in the city receives revenue from its subscribers as well as from advertisers.
- b) Advertising rates charged by the most profitable newspaper in the city are significantly higher than those charged by its competitors.
- c) Advertisers generally switch from the most widely circulated newspaper to another one only when the other one becomes the most widely circulated newspaper instead.
- ☒ d) The number of newspapers competing viably with the most profitable newspaper in the city has increased during the last two years.
- e) The circulation of the most profitable newspaper in the city is still greater than that of any of its competitors.

37. A study of native-born residents in Newland found that two-thirds of the children developed considerable levels of nearsightedness after starting school, while their illiterate parents and grandparents, who had no opportunity for formal schooling, showed no signs of this disability. If the above statements are true, which of the following conclusions is most strongly supported by them?

- ☒ a) The nearsightedness in the children is caused by the visual stress required by reading and other class work.
- b) One-third of the children are illiterate.
- c) People who are illiterate do not suffer from nearsightedness.
- d) Only literate people are nearsighted.
- e) Only people who have the opportunity for formal schooling develop nearsightedness.

38. Beautiful beaches attract people, no doubt about it. Just look at this city's beautiful beaches, which are among the most overcrowded beaches in the state. Which of the following exhibits a pattern of reasoning most similar to the one exhibited in the argument above?

- ☒ a) During warm weather my dog suffers from fleas more so than during cooler weather. Therefore, fleas must thrive in a warm environment.
- b) Children who are scolded severely tend to misbehave more often than other children. Hence if a child is not scolded severely that child is less likely to misbehave.
- c) Moose and bear usually appear at the same drinking hole at the same time of day. Therefore, moose and bear must grow thirsty at about the same time.
- d) Pesticides are known to cause anemia in some people. However, most anemic people live in regions where pesticides are not commonly used.
- e) This software program helps increase the work efficiency of its users. As a result, these users have more free time for other activities.



39. Smoking in bed has long been the main cause of home fires. Despite a significant decline in cigarette smoking in the last two decades, there has been no comparable decline in the number of people killed in home fires. Each one of the following statements, if true over the last two decades, helps to resolve the apparent discrepancy above EXCEPT:

- a) Compared to other types of home fires, home fires caused by smoking in bed usually cause relatively little damage before they are extinguished.
- b) An increasing number of people have been killed in home fires that started in the kitchen.
- c) Home fires caused by smoking in bed often break out after the home's occupants have fallen asleep. ★
- d) Smokers who smoke in bed tend to be heavy smokers who are less likely to quit smoking than are smokers who do not smoke in bed.
- e) Population densities have increased, with the result that one home fire can cause more deaths than in previous decades.

40. Video arcades, legally defined as video parlors having at least five video games, require a special city license and, in primarily residential areas such as Eastview, a zoning variance. The owners of the Video Zone, popular with Eastview teenagers, have maintained that their establishment requires neither an arcade license nor a zoning variance, because it is really a retail outlet. Which one of the following is an assumption upon which the argument of the Video Zone owners is based?

- a) Stores like the Video Zone perform an important social function.
- b) Retail establishments require no special licenses or zoning variances in Eastview.
- c) The existing Eastview zoning regulations are unconstitutionally strict.
- d) Many of the Video Zone's games were developed after the city's zoning laws were written.
- e) At no time are more than four video games in operation at the Video Zone.

41. No national productivity measures are available for underground industries that may exist but remain unreported. On the other hand, at least some industries that are run entirely by self-employed industrialists are included in national productivity measures. From the information given above, it can be validly concluded that:

- a) There are at least some industries run entirely by self-employed industrialists that are underground industries.
- b) There are at least some industries run entirely by self-employed industrialists that are not underground industries.
- c) There are at least some industries other than those run entirely by self-employed industrialists that are underground industries.
- d) No industries that are run entirely by self-employed industrialists operate underground.
- e) There are at least some underground industries for which national productivity measures are available.

42. Sally has never received a violation from the Federal Aviation Administration during her 16-year flying career. Sally must be a great pilot. Which of the following can be said about the reasoning above?

- a) The argument is built upon hidden assumptions.
- b) This is an example of an argument that is directed against the source of the claim rather than the claim itself.
- c) The definitions of the terms create ambiguity.
- d) The argument uses circular reasoning.
- e) The argument works by analogy.

43. All students who submit the admission form before the deadline receive a test slip. Hamza received a test slip. Which of the following conclusions is definitely valid? ✓

- a) Hamza submitted the admission form before the deadline.
- b) Hamza did not submit the admission form before the deadline.
- c) Hamza may or may not have submitted the admission form before the deadline. ✓
- d) Only Hamza received a test slip.
- e) No student submitted the form before the deadline.

Handwritten calculations and diagrams:

- A sequence of numbers: 18, 36, 60, 90, 120.
- Arrows indicating a progression from 18 to 36, 36 to 60, 60 to 90, and 90 to 120.
- Factorizations:  $18 = 3 \times 6$ ,  $36 = 6 \times 6$ ,  $60 = 6 \times 10$ ,  $90 = 6 \times 15$ .
- A circled '43' with an arrow pointing to the sequence.
- A circled '90' with an arrow pointing to the sequence.

## Section 7: Sequence and Series

44. 6, 24, 60, 120, \_\_\_\_

- a) 110

Handwritten calculations for Question 44:

- $6 \rightarrow 1 \times 2 \times 3$
- $120 \rightarrow 4 \times 5 \times 6$
- $120 + 90 = 210$

44. 6, 24, 60, 120, \_\_\_\_

- a) 110
- b) 210
- c) 220
- d) 180

✓ 210 ✓  
 $120 + 90 = 210$

$6 \rightarrow 1 \times 2 \times 3$   
 $24 \rightarrow 2 \times 3 \times 4$   
 $60 \rightarrow 3 \times 4 \times 5$

$120 \rightarrow 4 \times 5 \times 6$   
 $210 \rightarrow 5 \times 6 \times 7$

45. 17, 36, 53, 68, \_\_\_\_, 92

- a) 83
- b) 81
- c) 85
- d) 71

$19, 17$   
 $\uparrow 15 \uparrow 13$

$n^3 - n$

$6 \rightarrow 2^3 - 2$   
 $24 \rightarrow 3^3 - 3$   
 $60 \rightarrow 4^3 - 4$

$17 + 19, +17, +15, +13, +11$

$6^3 - 6 = 216 - 6 = 210$   
 $5^3 - 5 = 125 - 5 = 120$

$\frac{81}{11} = 7 \frac{4}{11}$

46. 7, 12, 9, 19, 13, \_\_\_\_

- a) 30
- b) 22
- c) 28
- d) 32

✓ 28 ✓  
 $0, 1, 2, 3, 4$

$7 + 5 = 12$   
 $12 - 3 = 9$   
 $9 + 10 = 19$

$19 - 6 = 13$   
 $13 + 15 = 28$

47. 1, 1, 2, 4, 7, \_\_\_\_

- a) 11
- b) 12
- c) 6
- d) 8

$+0, +1, +2, +3, +4$

$+5, -3, +10, -6, +15$

48. 1, 7, 17, 31, 49, 71, \_\_\_\_

- a) 92
- b) 91
- c) 97
- d) 93
- e) 98

$1 + 6 = 7$   
 $7 + 10 = 17$   
 $17 + 14 = 31$   
 $31 + 18 = 49$   
 $49 + 22 = 71$   
 $71 + 26 = 97$

$49$   
 $+ 22$   
 $\hline 71$   
 $+ 26$   
 $\hline 97$

$$\frac{102}{97}$$

49. 2, 6, 14, 26, 42, 62, \_\_\_\_  $2 + \textcircled{4} = 6 + \textcircled{8} = 14$

a) 92

b) 86

c) 72

d) 82

$$+4, +8, +12, +16, \dots$$

$$14 + \textcircled{12} = 26 + \textcircled{16} = 42 + \textcircled{20} = 62$$

50. 0, 1, 2, 4, 6, 9, 12, \_\_\_\_

a) 18

b) 14

c) 16

d) 15

$$+1, +1, +2, +2, +3, +3, +4, +4 = \textcircled{16}$$

$$62 + \textcircled{24} =$$

51. 1, 2, 6, 24, \_\_\_\_, 720

a) 510

b) 140

c) 360

d) 120

$$1!, 2!, 3!, 4!, \dots$$

$$5!$$

$$\checkmark \textcircled{120} \rightarrow 5!$$

$$1! = 1$$

$$2! = 2$$

$$3! = 3 \times 2 \times 1 = 6$$

$$4! = 4 \times 3 \times 2 \times 1 = \textcircled{24}$$

$$5! = 5 \times 4 \times 3 \times 2 \times 1$$

$$5 \times 4! = \textcircled{120}$$

52. 1, 3, 8, 19, 42, \_\_\_\_

a) 83

b) 91

c) 89

d) 64

$$2, 5, 11, 23, \dots$$

$$2, \textcircled{4+1}, 10+1=11$$

$$42 + \textcircled{47} =$$

$$\frac{47}{89}$$

53. 7, 9, 13, 19, \_\_\_\_

a) 31

b) 25

$$2, 4, 6, +8, \dots$$

a) 31

b) 25

c) 29

d) 27

54. 13, 15, 19, 25, 33, \_\_\_\_

a) 41

b) 49

c) 43

d) 38

55. 225, 336, 447, 558, 669, 7710

a) 556

b) 779

c) 557

d) 599

e) 558

56. 2520, 360, 60, 12, \_\_\_\_

a) 14

b) 3

c) 18

d) 13

57. 6, 9, 3, 8, 3, 7

a) 10

b) 7

c) 6

d) 8

58. 17, 11, 28, 39, 67, \_\_\_\_

a) 95

b) 58

c) 106

d) 96

2 4 6 8 +10

$$\begin{array}{r} 360 \\ 7 \overline{) 2520} \\ \underline{420} \\ 420 \end{array}$$

$$\frac{360}{6} = 60, \frac{60}{5} = 12$$
$$12 / 4 = 3$$

$$6 + 3 = 9, 9 - 6 = 3$$

$$5 + 3 = 8, 8 - 5 = 3$$

$$4 + 3 = 7$$

$$\begin{array}{r} + \\ 17 \\ 11 \\ 28 \\ \hline 56 \end{array}$$

$$5 \times 2 = 10$$
$$10 + 2 = 12$$

$$\begin{array}{r} 28 \\ 11 \\ \hline 39 \\ 67 \end{array}$$

11



c) 106

d) 96

$$5 \times 2 = 10 \\ 10 + 2 = 12$$

$$\begin{array}{r} 67 \\ \hline 106 \end{array}$$

59 (5, 10, 12), 6, 12, 14, 7, 14, 16

a) 8

b) 5

c) 9

d) 6

e) 7

$$(5, 10, 12), (6, 12, 14), (7, 14, 16)$$

$$-25, -25, -25$$

60. 75, 50, 90, 65, 105, \_\_\_\_

a) 185

b) 170

c) 80

d) 130

$$-25, +40, -25, +40, -25$$

$$+3, +5, +7, +9$$

61. 0, 3, 8, 15, 24, \_\_\_\_

a) 28

b) 25

c) 40

d) 35

$$+3, +5, +7, +9, +11$$

$$+2, +4, +6, +8 \checkmark$$

62. 13, 15, 19, \_\_\_\_, 43, 75, 139

a) 36

b) 48

c) 27

d) 42

$$+2, +4, +6, +8, +10$$

$$27 + 16$$

$$-2, \times 4, -2, \times 4, -2, \times 4$$

$$88 - 2 = 86$$

63. 4, 2, 8, 6, 24, 22, \_\_\_\_

a) 216

b) 256

c) 90

d) 200

e) 86

$$-2, \times 4, -2, \times 4, -2, \dots$$

$$4, 8, 16, 32, 64$$

$$\Rightarrow 2 \times 2 + 1$$



e) 86

4 8 16 32 64

64. 3, 7, 15, 31, 63, \_\_\_\_\_

a) 95

b) 111

c) 127

d) 129

e) 131

$$63 \times 2 + 1$$

$$3 \times 2 + 1$$

$$7 \times 2 + 1 = 15$$

$$15 \times 2 + 1 = 31$$

+4 +8 16 32 +64

65. 5, 9, 17, 33, 65, \_\_\_\_\_

a) 97

b) 111

c) 129

d) 131

e) 125

36-6 Check Point

$$16-4$$

$$5^2-5$$

$$n^2 - n$$

$$n(n-1)$$

$$n=1, n=2$$

$$n=3$$

$$4-2=2$$

0, 2, 6, 12, 20, 30, \_\_\_\_\_

2 4 6 8 10

3 9 27 72

-1 2 12 40 112 \_\_\_\_\_

7

288 284

~~1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100~~  
a) ~~280~~, b)  $28^2$ ,  $286$ ,  $288$ ,  $284$

$-1 \times 1, 1 \times 2, 3 \times 4, 5 \times 8, 7 \times 16, \dots$

$$a_n = (n-3) \times \underline{2^{n-1}}$$

$-1, 1, 3, 5, 7, \dots$

$$a_n = -1 + (n-1)2 = -1 + 2n - 2$$
$$a_n = 2n - 3$$